llmXive Automated Review of Function-Aware Fill-in-the-Middle as Mid-Training for Coding Agent Foundation Models

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper presents a novel mid-training strategy for coding agents, but several central claims rely on citations to works and models that do not currently exist in the public record, creating a significant risk of hallucinated baselines or results.

First, the methodology and experiments repeatedly reference "Gemini-3-Flash" (Abstract, Section 3.4, Section 4.1, Appendix B.8) as the teacher model for generating Chain-of-Thought rationales. The bibliography contains no entry for a "Gemini-3" model or a "Gemini-3-Flash" variant. While the authors may have access to internal or future models, citing a non-existent public model as a core component of the reproducibility pipeline is a factual error. The authors must verify the actual model version used (e.g., Gemini-1.5 Pro/Flash) and provide the correct citation, or explicitly state if the model is hypothetical.

Second, and more critically, the evaluation relies on "Terminal-Bench 2.0" and "SWE-Lego". The bibliography lists `terminalbench` (Merrill et al., 2026, arXiv:2601.11868) and `swelego` (Tao et al., 2026, arXiv:2601.01426). As of the current date, these papers and the associated benchmarks/pipelines are not publicly available. The dates (2026) and arXiv IDs suggest these are either future-dated hallucinations or placeholders. If these baselines are central to the paper's claim of "state-of-the-art" performance or "cross-pipeline" robustness (Section 4.2), the results cannot be verified or reproduced. The authors must either provide the actual, existing baselines used (e.g., Terminal-Bench 1.0, SWE-Gym, or SWE-Smith) and correct the citations, or acknowledge that these specific results are based on unreleased/future work which invalidates the current empirical claims.

Finally, the citation `ast-fim` (Gong et al., 2025) in Section 6 is also dated 2025. While less critical than the baselines, the pattern of citing 2025/2026 papers for standard benchmarks and methods suggests a systematic issue with the bibliography's accuracy. The authors should audit the entire reference list for future-dated or non-existent entries.

These issues are not merely stylistic; they undermine the verifiability of the paper's core experimental results. If the baselines (Terminal-Bench 2.0, SWE-Lego) are fabricated, the paper's primary contribution claims are unsupported.

Action items.

- **[writing]** The paper presents a novel mid-training strategy for coding agents, but several central claims rely on citations to works and models that do not currently exist in the public record, creating a significant risk of hallucinated baselines or results. First, the methodology and experiments repeatedly reference "Gemini-3-Flash" (Abstract, Section 3.4, Section 4.1, Appendix B.8) as the teacher model for generating Chain-of-Thought rationales. The bibliography contains no entry for a "Gemini-3" model

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure effectively illustrates the conceptual analogy and training pipeline, but the results chart suffers from ambiguous x-axis labels that fail to identify the specific models used, and the legend abbreviations are not explicitly defined in the chart itself.

Figure 2

Figure 2 effectively visualizes the scoring components for the calculator example, but the caption contains garbled text regarding the FIM calculation and missing equation references. Additionally, the legend in panel (b) fails to define one of the five segments in the inferability score bar.

Figure 3

The bar chart effectively displays repository counts, but the caption's textual summary contradicts the visual data regarding dominant categories, and the x-axis labels are poorly formatted with excessive overlap.

Figure 4

Figure 4 is a clear and well-constructed donut chart that effectively visualizes the license distribution of the corpus. The legend is comprehensive, explicitly listing all categories and their percentages, and aligns perfectly with the caption's description of the data aggregation.

Figure 5

The figure clearly displays pass rates stratified by patch shape, but it omits error bars despite claiming to show means over three runs, and the x-axis labels are unnecessarily split across lines.

Figure 6

The figure presents stacked bar charts of outcome distributions, but the caption's claim of averaging over three runs contradicts the integer counts shown, and the 'No-patch' segment lacks a numerical label.

Figure 7

The figure effectively displays pass rates for the relevant categories, but it misleadingly includes a 'multi-file' bar with n=0 despite the caption stating such tasks do not exist in the dataset. Additionally, the x-axis labels are unnecessarily split across lines.

Action items.

- [**science**] Figure 1: The bar chart x-axis labels (7B, 14B, 8B) are ambiguous and do not match the caption's specific model names (Qwen2.5-Coder-Instruct, Qwen3), making it impossible to identify which model corresponds to the 8B bar without external knowledge.
- [**writing**] Figure 1: The legend in the bar chart uses 'Verified' and 'Lite' labels, but the caption specifies these refer to 'SWE-Bench-Verified' and 'SWE-Bench-Lite'; the chart should use the full names or the caption should explicitly define the abbreviations.
- [**writing**] Figure 2 caption: The text 'yielding FIM 0.22 = 0.08' is garbled and mathematically incoherent; it likely intends to state that the FIM score is 0.22 and the threshold *au* is 0.08.
- [**writing**] Figure 2 caption: The text references 'Eq. ;' for both complexity and inferability scores, indicating missing equation numbers.
- [**science**] Figure 2b: The stacked bar for the inferability score (I) contains five segments, but the legend below only provides four labels (caller, callee, sig, doc, class), leaving one segment undefined.
- [**science**] Figure 3: The caption states the corpus is dominated by 'reference implementations, scientific computing, and small frameworks,' but the chart labels 'From Scratch' (271) and 'Educational' (139) are the largest categories, while 'Small Frameworks' (118) is smaller than 'Educational' and 'Scientific Computing' (131). The caption's summary does not accurately reflect the data shown.
- [**writing**] Figure 3: The x-axis labels are rotated at a steep angle, causing significant overlap and making the text difficult to read (e.g., 'Visualization and Games' and 'Data Processing' are crowded).
- [**science**] Figure 5: The caption states the data is 'run-means over three evaluation runs per checkpoint,' but the figure lacks error bars or any indication of variance/standard deviation, which is standard for reporting means over multiple runs.
- [**writing**] Figure 5: The x-axis labels ('single-func single-file', 'multi-func single-file', 'multi-file') are split across two lines, reducing readability; consider using a single line or adjusting font size.
- [**science**] Figure 6: The stacked bars sum to ~500 (n=500), but the caption states 'averaged over three runs'; averaging counts across runs without normalizing by the number of tasks per run is statistically invalid and misleading.
- [**writing**] Figure 6: The red 'No-patch' segment is visually present in the left bar but lacks a corresponding numerical label, unlike all other segments.

- [**science**] Figure 7: The 'multi-file' category is plotted with n=0 and 0.0% pass rate, but the caption explicitly states 'Lite contains no multi-file tasks'; plotting a non-existent category with zero values is misleading and should be removed.
- [**writing**] Figure 7: The x-axis label 'single-func single-file' is split across two lines, creating unnecessary visual clutter; this should be formatted on a single line or wrapped more cleanly.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper is generally well-written and avoids excessive in-group slang, but it relies heavily on a custom mathematical notation system for the function-aware FIM selection pipeline (Section 3.2) that is introduced with insufficient explicit definition for a reader outside the immediate subfield of code-aware LLM training.

Specifically, the paper introduces a suite of symbols ($\hat{H}$, $\hat{I}$, Δ, ρ, ϕ, $\mathcal{E}$, Coup, $\overline{\text{Jacc}}$) in rapid succession within Equations 1–5. While the authors provide definitions in the surrounding text or appendices, the density of the notation creates a high cognitive load. A competent adjacent-field PhD (e.g., in NLP or software engineering) would likely pause to verify whether ϕ refers to the golden ratio, Δ to a Laplacian, or ρ to a correlation coefficient, as these are standard usages in other fields. The paper assumes the reader will infer the specific, non-standard definitions from the immediate context, which is a barrier to entry.

Additionally, the acronyms for the post-training pipelines (R2E-Gym, SWE-Smith, SWE-Lego) are treated as common knowledge. While they are cited, a brief operational description (e.g., "R2E-Gym, a benchmark for procedural reasoning...") would significantly improve accessibility for readers from adjacent fields who may not be tracking the latest agent-specific benchmarks.

The definitions are present, but they are often buried in the text following the equation or in the appendix, rather than being integrated into the equation's immediate context or the sentence introducing the symbol. This forces the reader to flip back and forth or parse dense paragraphs to reconstruct the meaning of the variables.

To improve accessibility, the authors should:

1. Explicitly define every custom symbol (ϕ, Δ, ρ, etc.) at its first occurrence, ideally within the sentence introducing the equation or as a "where" clause immediately following the equation.
2. Clarify the range and units of the normalized scores (C terms) to ensure the reader understands the scale of the values.
3. Provide a one-sentence operational definition for the named pipelines (R2E-Gym, etc.) upon first mention.

These are minor text edits that will significantly lower the barrier to entry for a broader technical audience without diluting the technical precision of the work.

Action items.

- [**writing**] Explicitly define every custom symbol (ϕ, Δ, ρ, etc.) at its first occurrence, ideally within the sentence introducing the equation or as a "where" clause immediately following the equation.
- [**writing**] Clarify the range and units of the normalized scores (C terms) to ensure the reader understands the scale of the values.
- [**writing**] Provide a one-sentence operational definition for the named pipelines (R2E-Gym, etc.) upon first mention. These are minor text edits that will significantly lower the barrier to entry for a broader technical audience without diluting the technical precision of the work.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper's argument structure is logically sound and internally consistent. The central thesis—that the structural isomorphism between function calls and agent steps justifies a function-aware FIM mid-training stage—is clearly stated in the Introduction (Section 2) and Method (Section 3), and the experimental design in Section 4 directly tests this hypothesis.

The logical flow from premises to conclusions holds:

1. **Premise:** Agent steps (context → action → observation → continuation) mirror function calls (context → call → return → usage).
2. **Method:** The authors construct a training objective (FIM) that explicitly targets this structure using program dependency graphs and complexity/inferability scores.
3. **Evidence:** Table 1 (Main Results) and Table 2 (Capability Preservation) show that models trained with this objective outperform baselines on agent benchmarks and recover capabilities eroded by standard post-training.
4. **Conclusion:** The gains are attributed to the structural prior, a claim supported by the ablation studies (Table 3) which isolate the FIM structure from CoT distillation and show that the effect persists across different base models and post-training pipelines.

There are no contradictions between sections. The limitations section (Section 7) accurately scopes the claims made in the conclusion, explicitly noting the Python-only corpus and the single non-Qwen base model configuration, which aligns perfectly with the experimental setup described in Section 4. The numerical values in the text (e.g., the +2.8 and +3.0 gains in the Abstract and Introduction) match the data presented in Table 1. The ablation text in Section

4.3 correctly interprets the data in Table 3, identifying the “Full” selection algorithm as the dominant factor, which is consistent with the table’s ranking.

The argument does not overreach; causal language is appropriately hedged where the evidence is correlational (e.g., “suggests,” “evidence for”), and the distinction between the structural analogy and the empirical transfer is maintained throughout. No logical gaps, non-sequiturs, or internal contradictions were found.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper demonstrates strong internal consistency and includes a dedicated Limitations section that appropriately scopes many claims. However, the Abstract and Conclusion contain rhetorical overreach where the language implies broader generalization than the specific experimental evidence supports.

1. **Cross-Domain Generalization Scope:** The Abstract and Conclusion assert “consistent gains” on non-coding tool-use benchmarks (τ-bench, BFCL). This claim is supported only by the 14B model results in Table 4. The 7B model’s performance on these specific cross-domain benchmarks is not reported. Claiming consistency across the tested regimes without the 7B data for these metrics overstates the evidence. The claim should be explicitly limited to the 14B configuration.
2. **Base-Model Family Generalization:** The Conclusion states the recipe works “across... base-model family.” The sole evidence is the Qwen3-8B + SWE-Lego experiment. As the authors correctly note in Section 4.2, this varies both the base model and the post-training pipeline simultaneously, making it a confounded comparison. While the body text hedges this, the Conclusion presents it as a definitive cross-family result. This should be narrowed to reflect the single, confounded data point.
3. **Magnitude of Recovery:** The Abstract claims “consistent gains” on non-coding benchmarks. Table 4 shows that while LiveCodeBench sees a large recovery (+11.1), gains on OJBench (+1.94) and FullStackBench-EN (+0.53) are marginal. The term “consistent” is misleading given this variance. The phrasing should be adjusted to reflect the mixed magnitude of recovery.

These issues are fixable by tightening the Abstract and Conclusion to match the specific scope of the data presented in the results tables.

Action items.

- [**writing**] Abstract claims ‘consistent gains’ on non-coding benchmarks, but Table 4 shows this is only tested on 14B. The 7B cross-domain results are missing. Scope the claim to the

14B configuration or add 7B data.

- **[writing]** Conclusion claims gains 'across base-model family,' but evidence is a single confounded Qwen3-8B+SWE-Lego run (Section 4.2). Narrow to 'a single non-Qwen2.5 configuration' to match the evidence.
- **[writing]** Abstract states 'consistent gains' on tool-use benchmarks, yet Table 4 shows negligible gains on OJBench (+1.94) and FullStackBench-EN (+0.53). Qualify 'consistent' to reflect the mixed magnitude of recovery.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The paper presents a method for mid-training coding agents using function-aware fill-in-the-middle (FIM) on a corpus of 968 open-source Python repositories. From a safety and ethics perspective, the work is low-risk.

The authors explicitly address the primary safety concerns in the NeurIPS Checklist (included in the source):

1. **Data Provenance & Licensing:** The corpus is constructed from public GitHub repositories. The authors state in Section 3.2 and Appendix A.1 that they performed decontamination against SWE-Bench and verified licenses. Appendix A.1 includes a license inventory (Figure 3) showing the corpus is dominated by permissive licenses (MIT, Apache-2.0, BSD), with all others being research-permissive. There is no indication of scraping against Terms of Service or using non-permissive data.
2. **Human Subjects:** The paper correctly identifies that no human subjects were involved. The chain-of-thought rationales are generated by an LLM (Gemini-3-Flash), not humans, so IRB/consent is not applicable (Checklist Item 14).
3. **Dual-Use & Misuse:** The method improves coding agent capabilities. While improved coding agents can theoretically be used for malicious purposes (e.g., writing malware), the paper does not introduce a novel capability that significantly lowers the barrier to such harm compared to existing base models (Qwen2.5-Coder, Qwen3). The authors acknowledge in the Checklist (Item 11) that the released models do not introduce generative capabilities for harmful content beyond the underlying base models. The "Broader Impacts" section is noted as missing in the Checklist (Item 10), but the authors justify this by stating the work is incremental and risks are comparable to the base models. Given the nature of the contribution (a training technique for existing code models) and the explicit license compliance, a dedicated broader impacts essay is not a fatal omission, and the risk profile is consistent with standard code-generation research.

4. **Vulnerability Disclosure:** The paper evaluates on standard benchmarks (SWE-Bench) and does not disclose operational exploits or vulnerabilities in live systems.

No specific, nameable safety risks or missing disclosures were found that would prevent publication. The paper adheres to the NeurIPS Code of Ethics regarding data licensing and human subjects.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling hypothesis regarding the structural isomorphism between function calls and agent steps, supported by a detailed mid-training pipeline. The experimental design is generally robust, particularly the use of multiple seeds (n=3) for main results and the inclusion of ablation studies. However, three specific design gaps prevent the evidence from fully isolating the claimed mechanisms.

First, the claim of cross-base-model transfer (Table 1, Qwen3-8B) is confounded. The comparison varies both the base model (Qwen2.5 vs. Qwen3) and the post-training pipeline (R2E-Gym/SWE-Smith vs. SWE-Lego). The observed gain could be entirely attributable to the SWE-Lego pipeline's specific data distribution or hyperparameters rather than the FIM prior. To support the claim that the prior transfers across families, the authors must run the FIM-midtrained Qwen3-8B with one of the original pipelines (R2E-Gym or SWE-Smith) to isolate the variable of interest.

Second, the ablation study on function selection (Table 3, Block B) lacks a control for hyperparameter sensitivity. The "Full" configuration (PDG + H + I) is compared against partial variants, but all use the same fixed threshold $\tau = 0.08$. It is plausible that the "Full" score distribution simply aligns better with this specific threshold, whereas the partial variants might outperform if their thresholds were re-tuned. A sensitivity analysis or a grid search over thresholds for the partial baselines is required to confirm the superiority of the combined score itself.

Third, the claim that the "function-call inductive bias" drives cross-domain gains (Section 4.3) is not fully isolated from a general regularization effect. The mid-training corpus is Python-only, and the baseline (post-training only) suffers from overfitting to SWE-Bench. The improvement on τ-bench and BFCL could result from the mid-training stage simply acting as a regularizer that prevents catastrophic forgetting, rather than installing a specific structural prior. To rule this out, the authors should include a control run using a random-span FIM corpus of equivalent size and complexity (without the function-aware selection logic). If the random-span variant yields similar cross-domain recovery, the specific "function-aware" design is not the primary driver.

Action items.

- **[writing]** The paper presents a compelling hypothesis regarding the structural isomorphism between function calls and agent steps, supported by a detailed mid-training pipeline. The experimental design is generally robust, particularly the use of multiple seeds (n=3) for main results and the inclusion of ablation studies. However, three specific design gaps prevent the evidence from fully isolating the claimed mechanisms. First, the claim of cross-base-model transfer (Table 1, Qwen3-8B) is confounded. The

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical reporting in this paper is generally transparent regarding the use of multiple seeds (N=3) for the primary experiments, which is a positive adherence to ML norms. However, there are two specific areas where the reporting of uncertainty and inferential claims requires tightening to ensure the numbers mean what the text claims.

First, in **Table 1 (Section 4.2)**, the authors report means and standard deviations for their reproduced baselines and proposed method, but the rows labeled “officially reported” (e.g., R2E-Gym official: 19.00 $\pm$ 1.00) present single point estimates without standard deviations. While the text notes these are quoted from other publications, presenting them alongside results with explicit error bars creates an asymmetry in uncertainty reporting. If the original papers did not report variance, the authors should either omit the error bars for their own results in those specific comparison rows (to avoid false precision) or explicitly annotate that the baseline values are single-run estimates, preventing readers from inferring a statistical significance test was performed against a fixed number.

Second, the paper makes strong claims about “consistent gains” and “significant” improvements (e.g., Section 4.3, Section 5) based on differences calculated from only **three independent seeds**. While reporting the mean and standard deviation is good practice, the text implies statistical significance (e.g., “mid-training improves... by +2.80”) without performing or reporting a formal hypothesis test (such as a paired t-test or Wilcoxon signed-rank test) on the seed-level data. With N=3, the power to detect significance is low, and the observed differences could easily be due to random seed variance. The authors should either run the appropriate statistical tests on the per-seed results and report the p-values (or 95% confidence intervals for the difference) to substantiate claims of improvement, or soften the language to describe the results as “observed improvements” without invoking statistical significance. This is a reporting fix (writing) as the raw per-seed data is presumably available to the authors.

Action items.

- **[writing]** Table 1 (sec/4_experiments.tex) reports mean $\pm$ SD over three seeds for most rows, but the ‘officially reported’ baselines (e.g., R2E-Gym official) list single point estimates (e.g., 19.00) without uncertainty bands. To ensure fair statistical comparison, either report the SD for these baselines if available, or explicitly state that these are single-run citations and treat comparisons against them as descriptive rather than inferential.

- **[writing]** Section 4.3 (Table 2) and Section 5 (Table 3) report mean differences (Δ) and recovery rate improvements (e.g., +4.0 pp) derived from three seeds. However, no hypothesis tests (e.g., paired t-tests or Wilcoxon signed-rank) are performed to determine if these gains are statistically significant beyond the observed variance. Given the small N=3, report the p-values or confidence intervals for these differences to support claims of 'consistent gains' or 'significant' improvements.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The paper is generally well-structured and the argument flows logically from the motivation of the function-call/agent-step isomorphism to the method and results. However, there are specific instances where sentence construction impedes immediate comprehension, requiring the reader to re-parse or untangle complex clauses.

In Section 4.2, the discussion of the Qwen3-8B results contains a long, convoluted sentence that attempts to qualify the experimental design while stating the conclusion. The main point—that the result is evidence of generalizability but not a guarantee—is buried at the end of a clause-heavy sentence. Splitting this into two distinct sentences would clarify the causal link between the experimental confound and the interpretation.

Similarly, the final paragraph of Section 4.3 suffers from a garden-path structure. The sentence begins with a causal clause, moves to a mechanism description, and ends with a relative clause that awkwardly attaches to the entire preceding thought rather than a specific noun. This forces the reader to backtrack to understand what constitutes the "direct evidence." A rewrite that separates the mechanism from the evidentiary claim would resolve this friction.

In Section 5.2, the prose becomes slightly repetitive in the final paragraph, using "slice" twice in close proximity to describe the same concept. While not a grammatical error, this redundancy slows the reading pace. A more concise phrasing would improve the flow.

Finally, the opening sentence of Section 3.4 ("For each selected target we run a three-stage pipeline") is functional but weak as a topic sentence. It describes an action rather than stating the paragraph's purpose. Strengthening this to explicitly name the stages or the goal of the pipeline would better orient the reader immediately.

These issues are minor and do not obscure the scientific contribution, but addressing them would ensure the prose moves as smoothly as the logic.

Action items.

- **[writing]** The paper is generally well-structured and the argument flows logically from the motivation of the function-call/agent-step isomorphism to the method and results. However, there are specific instances where sentence construction impedes immediate comprehension, requiring the reader to re-parse or untangle complex clauses. In Section 4.2, the discussion of the Qwen3-8B results contains a long, convoluted sentence that attempts to qualify the experimental design while stating the conclusion. The m